Exploring the structure of policies produced by Markov Decision Processes for problems relating to redundant systems

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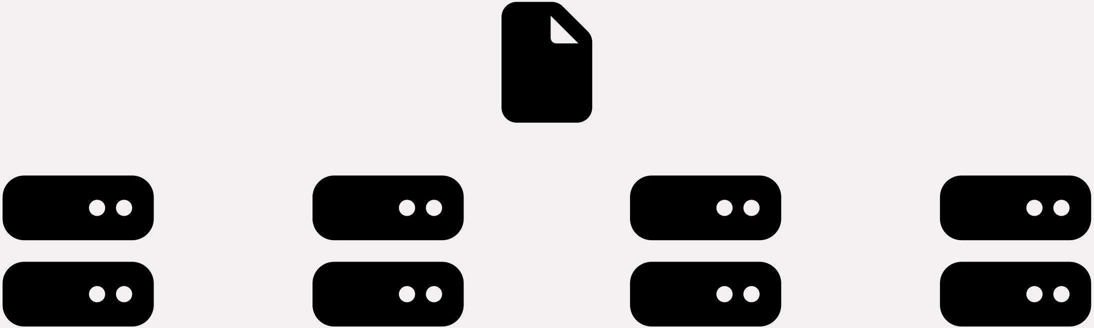
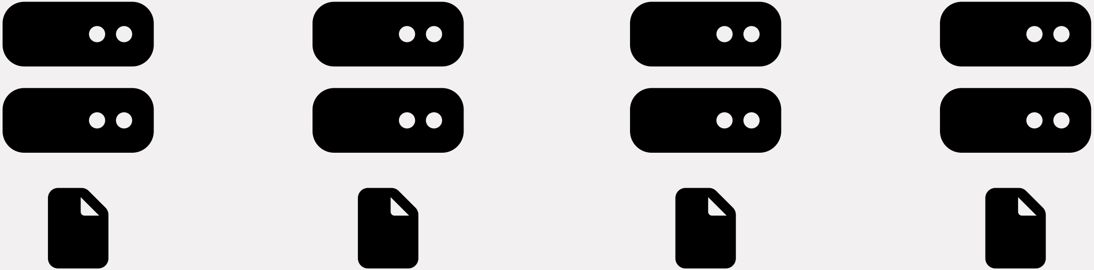
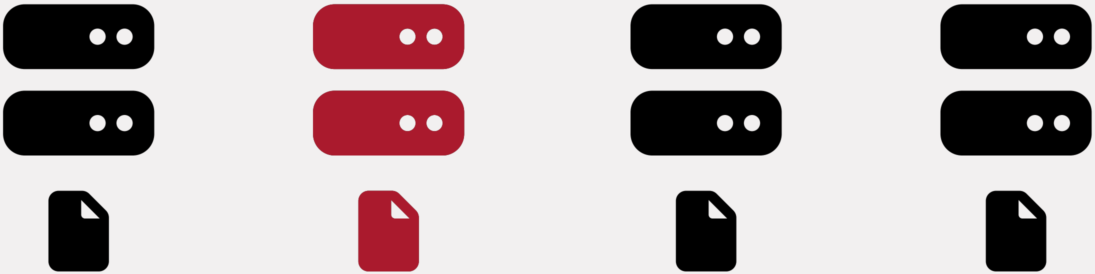
Motivation: erasure coding



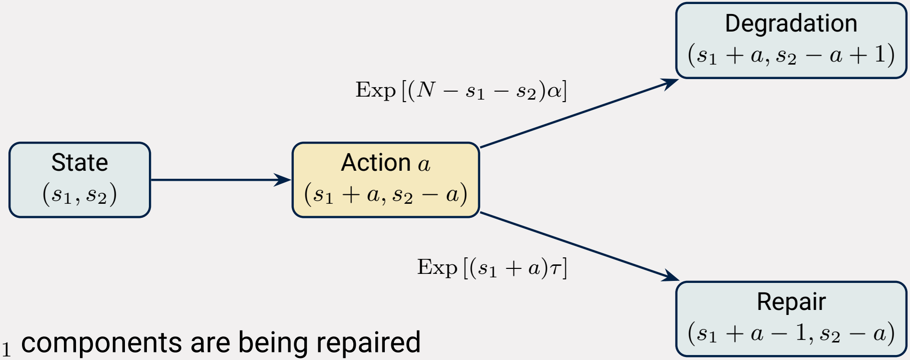
Motivation: erasure coding



Motivation: erasure coding



The Model

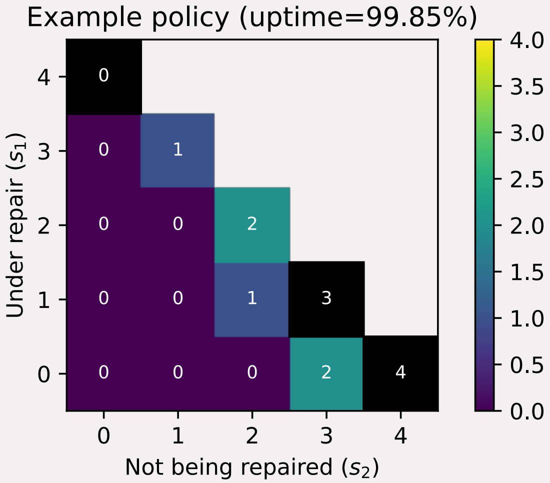


- s_1 components are being repaired
- s_2 components are broken but not being repaired
- $s_0 = N - s_1 - s_2$ components are healthy components
- We start repairing a components

From actions to a policy

$$\begin{aligned} \pi : \mathcal{S} &\longrightarrow \mathcal{A} \\ s &\longmapsto \pi(s) \end{aligned}$$

We will restrict ourselves to policies that are Markovian and deterministic



Which policy should we choose?

$$r(s,a) = -(s_1 + a)r - p\mathbb{1}_{\{s_0 < k\}}$$

Repair cost

$$-(s_1 + a)r$$

Repairing components is expensive

Downtime penalty

$$-p\mathbb{1}_{\{s_0 < k\}}$$

Downtime is very expensive

Optimising over time

Discounted reward

$$\mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(S^{(t)}, A^{(t)}) \right] \qquad \gamma \in (0, 1)$$

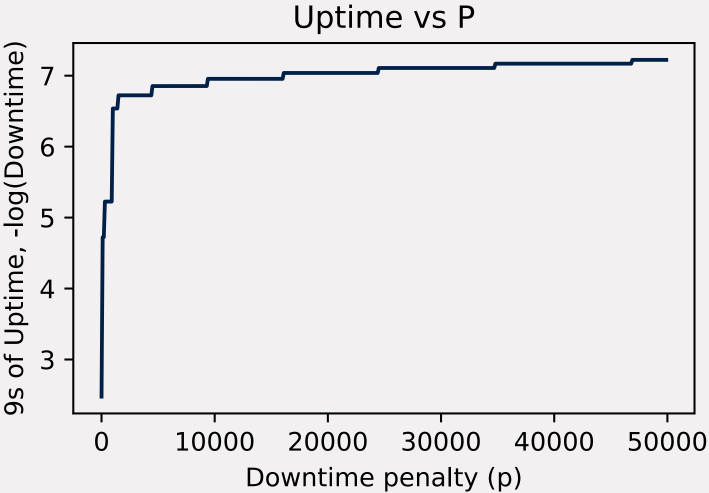
Rewards far in the future matter less
Requires a choice of γ

Long-run average reward

$$\limsup_{T \to \infty} \frac{1}{T} \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} r(S^{(t)}, A^{(t)}) \right]$$

Optimises steady state performance
Ignores rewards in transient states

Downtime penalty



Downtime penalty

Uptime monotonicity

Let $p_1 < p_2$, and let π_1, π_2 be corresponding optimal policies. Then

$$\text{Uptime}(\pi_1) \leq \text{Uptime}(\pi_2)$$

This means we can binary search for uptime

Action targets

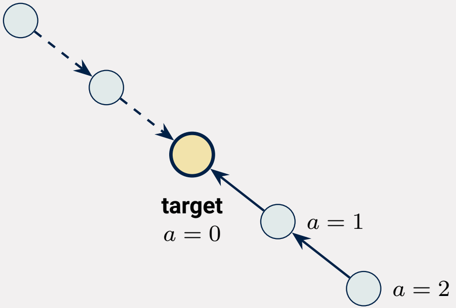
Let R be a recurrent class under an optimal policy π.

Action-target

If:

- $s = (s_1, s_2) \in R$
- a is an optimal action for s
- $0 \leq x \leq a$

Then: $a - x$ is an optimal action for $(s_1 + x, s_2 - x)$



Damage function

Damage structure of the recurrent class

In the recurrent class R, there is a unique least damaged state and it will have form $(0, \beta)$.
Moreover,
$$\{s_1 + s_2 \ : \ (s_1, s_2) \in R\} = \{\beta, \beta + 1, \ldots, N\}.$$

So 100% uptime is impossible

Conjectures

All of the following have been numerically tested across 11,760 problems.

1. **Zeros after the action target** If $x > a$ then 0 is an optimal action for $(s_1 + x, s_2 - x)$ for recurrent states on the same diagonal.
2. **Action monotonicity in the downtime penalty** If $p_1 < p_2$, then $\pi_1(s) \leq \pi_2(s)$ for any state belonging to both recurrent classes.
3. **Gravity of the recurrent class** If $(s_1, s_2) \in R, \quad 0 \leq s_1' \leq s_1$ then $(s_1', s_2) \in R$

If true then the optimal policy and recurrent class would admit an $O(N)$ description, despite the $O(N^2)$ state space.

Number of policies

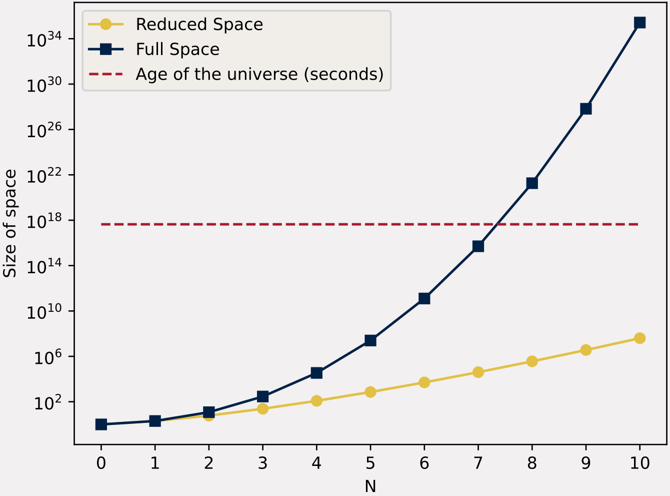
Before

$$\prod_{k=1}^{N+1} k!$$

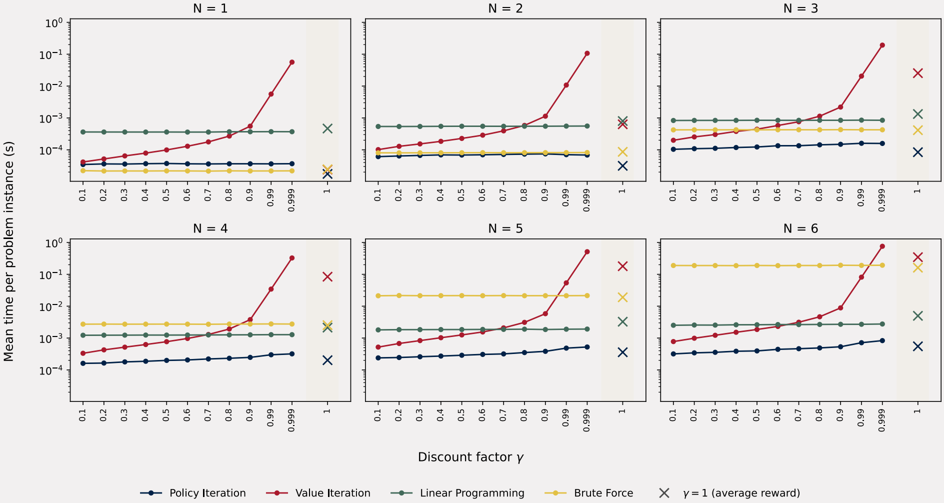
After

$$(N+1)!$$

Number of policies



Brute force


N = 1
N = 2
N = 3
N = 4
N = 5
N = 6
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 0.99 0.999 1
Mean time per problem instance (s)
Discount factor γ
Policy Iteration
Value Iteration
Linear Programming
Brute Force
$\gamma = 1$ (average reward)

Model limitations

- Independence of events
- Distribution of failures and repairs
- Homogeneity of components
- Recoverability

Any Questions?

Monotonicity of uptime in cost ratio: Dicounted

Let $p_1 < p_2$ and let π_1, π_2 be the respective optimal policies and define

$$C(s;\pi,\gamma) := \mathbb{E}\left(\sum_{t=0}^{\infty} r\gamma^t \left(S_1^{(t)} + \pi\left(S^{(t)}\right)\right) \middle| S_0 = s\right)$$
$$D(s;\pi,\gamma) := \mathbb{E}\left(\sum_{t=0}^{\infty} \gamma^t \mathbb{1}_{\{S_1^{(t)}+S_2^{(t)}=N\}} \middle| S_0 = s\right)$$

Each optimal policy minimises total discounted cost $C(s;\pi_i,\gamma) + p_i\, D(s;\pi_i,\gamma)$, so

$$C(\pi_1) + p_1 D(\pi_1) \leq C(\pi_2) + p_1 D(\pi_2) \quad C(\pi_2) + p_2 D(\pi_2) \leq C(\pi_1) + p_2 D(\pi_1)$$
$$\implies -(p_2 - p_1) D(\pi_1) \leq -(p_2 - p_1) D(\pi_2) \implies D(s;\pi_2,\gamma) \leq D(s;\pi_1,\gamma) \quad \forall s \in \mathcal{S}$$

Monotonicity of uptime in cost ratio: Undiscounted

Let $p_1 < p_2$ and let π_1, π_2 be the respective optimal policies and define

$$C(s;\pi) := \lim_{N\to\infty} \frac{1}{N}\mathbb{E}\left(\sum_{t=0}^{N-1} r\left(S_1^{(t)} + \pi\left(S^{(t)}\right)\right) \middle| S_0 = s \right)$$

$$D(s;\pi) := \lim_{N\to\infty} \frac{1}{N}\mathbb{E}\left(\sum_{t=0}^{N-1} \mathbb{1}_{\{S_1^{(t)}+S_2^{(t)}=N\}} \middle| S_0 = s \right)$$

Each optimal policy minimises total long-run average cost $C(s;\pi_i) + p_i\, D(s;\pi_i)$, so

$$C(\pi_1) + p_1 D(\pi_1) \leq C(\pi_2) + p_1 D(\pi_2) \quad C(\pi_2) + p_2 D(\pi_2) \leq C(\pi_1) + p_2 D(\pi_1)$$
$$\implies -(p_2 - p_1)D(\pi_1) \leq -(p_2 - p_1)D(\pi_2) \implies D(s;\pi_2,\gamma) \leq D(s;\pi_1,\gamma) \quad \forall s \in \mathcal{S}$$

Actions on a diagonal: Undiscounted

Lemma: Let π be the optimal policy with an induced recurrent class R. Let $s = (s_1, s_2) \in R, a := \pi(s)$. IF $0 \leq x \leq a$ and $s' := (s_1 + x, s_2 - x) \in R$ THEN $\pi(s') = a - x$

First note that the reward and probability functions depend only on the post-action state hence we can simplify the optimality equations to

$$
\begin{aligned}
g + h(s) &= \max_{b \in B(s)} f(b) \qquad \text{where} \\
B(s) &= \{(s_1 + b, s_2 - b) : 0 \leq b \leq s_2\} \\
f(b) &= -rb_1 - p\mathbb{1}_{\{N - b_1 - b_2 \leq k\}} + \sum_j p(j|b)h(j)
\end{aligned}
$$

Note that $B(s') \subseteq B(s)$ so $\max_{b \in B(s')} f(b) \leq \max_{b \in B(s)} f(b)$
Since $(s_1 + a, s_2 - a) \in B(s')$ we have that
$f(s_1 + a, s_2 - a) \leq \max_{b \in B(s')} f(b) \leq \max_{b \in B(s)} f(b) = f(s_1 + a, s_2 - a)$
So the post-action state for s' and s is the same

Actions on a diagonal: Discounted

Lemma: Let π be the optimal policy with an induced recurrent class R.
Let $s = (s_1, s_2) \in R, a := \pi(s)$. IF $0 \leq x \leq a$ and $s' := (s_1 + x, s_2 - x) \in R$
THEN $\pi(s') = a - x$

First note that the reward and probability functions depend only on the post-action state hence we can simplify the optimality equations to

$$
\begin{aligned}
v(s) &= \max_{b \in B(s)} f(b) \qquad \text{where} \\
B(s) &= \{(s_1 + b, s_2 - b) : 0 \leq b \leq s_2\} \\
f(b) &= -rb_1 - p\mathbb{1}_{\{N - b_1 - b_2 \leq k\}} + \gamma \sum_j p(j|b)v(j)
\end{aligned}
$$

Note that $B(s') \subseteq B(s)$ so $\max_{b \in B(s')} f(b) \leq \max_{b \in B(s)} f(b)$
Since $(s_1 + a, s_2 - a) \in B(s')$ we have that
$f(s_1 + a, s_2 - a) \leq \max_{b \in B(s')} f(b) \leq \max_{b \in B(s)} f(b) = f(s_1 + a, s_2 - a)$
So the post-action state for s' and s is the same

0s on diagonals

If $x > a$ in the above lemma then we still have that $B(s') \subseteq B(s)$ so still have that $f(s') = \max_{b \in B(s')} f(b) \leq \max_{b \in B(s)} f(b) = f(s_1 + a, s_2 - a)$. We then define $F(x) = f(s_1 + x, s_2 - x)$, if we assume that F is monotonically decreasing on $\{a, a+1, \ldots, s_2\}$ (which is unproved) then the post-action state for s' and s is the same so $\pi(s_1 + x, s_2 - x) = 0$

0s on diagonals: the top diagonal

Take $s = (0, N)$, so that $F(x) = f(x, N - x)$ and $d(s) = N$. Here $s_0 = 0$, so nothing can degrade and the only transition out of $(x, N - x)$ is a repair into $(x - 1, N - x)$. Writing Δ for the uniformisation constant,

$$F(x) = -(rx + p) + \gamma \Delta x \tau \, v(x - 1, N - x) + \gamma \, (1 - \Delta x \tau) \, v(x, N - x)$$

Lemma: if $d(s) = d(s')$ and $s_1 \leq s_1'$ then $v(s) \geq v(s')$

$B(s') \subseteq B(s)$, so $\max_{b \in B(s')} f(b) \leq \max_{b \in B(s)} f(b)$.

Now suppose $\pi(0, N) = 0$, i.e. $v(0, N) = F(0)$. Then

- $F(x) \leq F(0)$ for every x, as $(x, N - x) \in B(0, N)$
- $F(0) = \dfrac{-p}{1 - \gamma}$, as $(0, N)$ has both rates zero, so $F(0) = -p + \gamma F(0)$

0s on diagonals: the top diagonal

Lemma: if $\pi(0, N) = 0$ then 0 is an optimal action for $(x, N-x) \quad \forall x$

Suppose F is not monotonically decreasing and let i be the largest index with $F(i) < F(i+1)$; then $i \geq 1$ as $F(1) \leq F(0)$, and F is monotonically decreasing on $\{i+1, \ldots, N\}$, so

$$v(i, N-i) = v(i+1, N-i-1) = F(i+1)$$

Rearranging the equation for F at $i+1$ and at i

$$\begin{aligned}
(1-\gamma)F(i+1) + p &= (i+1)\Big[\gamma\Delta\tau\big(v(i, N-i-1) - F(i+1)\big) - r\Big] \\
(1-\gamma)F(i+1) + p &> \ \ i \qquad \Big[\gamma\Delta\tau\big(v(i, N-i-1) - F(i+1)\big) - r\Big]
\end{aligned}$$

So the bracket is positive, giving $F(i+1) > \dfrac{-p}{1-\gamma} = F(0)$, a contradiction.

Hence F is monotonically decreasing, so its maximum over $B(x, N-x)$ is attained at $(x, N-x)$, i.e. $v(x, N-x) = F(x)$.

0s on diagonals: $N = 2$

Corollary: for $N = 2$ conjecture 1 holds at every state

$$\begin{array}{lll} 0 & & \\ 0 & \{0,1\} & \\ 0 & \{0,1\} & \{0,1,2\} \end{array}$$

Only place it could fail is on the diagonal (every other diagonal has no more than 1 state that can vary). The above lemma shows it is true on this upper diagonal.

Cancellation on a diagonal

Consider a model with cancellation i.e. $-s_1 \leq a \leq s_2$

Lemma: Let π be the optimal policy with an induced recurrent class R. Let $s = (s_1, s_2) \in R, a := \pi(s)$. IF $s' := (s_1 + x, s_2 - x) \in R$ THEN $\pi(s') = a - x$

We use f as before but now define B as follows

$$B(s) = \{(s_1 + b, s_2 - b) : -s_1 \leq b \leq s_2\}$$

Since $B(s)$ depends only on $s_1 + s_2$, $B(s) = B(S')$
This immediately gives that $\max_{b \in B(s')} f(b) = \max_{b \in B(s)} f(b)$
So the post-action state for s' and s is the same

This gives us our $O(N)$ structure we've been looking for

Happiness exists but is unique

Lemma: $(0, \beta)$ is the unique state with minimal damage function ($d(s) = s_1 + s_2$)

Let $s \in R$ such that $d(s)$ is minimal. We will show that $s_1 = 0$. This is enough to conclude that $s_2 = \beta$ (as defined above), and hence that s is unique.

Case 1: $s_1 + a > 0$

The transition rate to repair is $(s_1 + a)\tau > 0$,
so the transition to $(s_1 + a - 1,\ s_2 - a)$ has positive probability,
this contradicts the minimality of d

Case 2: $s_1 + a = 0 \implies s_1 = 0,\ a = 0.$

Damage function

Lemma: $\{d(s) : s \in R\} = \{\alpha_R^{\min}, \alpha_R^{\min} + 1, \ldots, N\}$

$\subseteq$ by previous result.
Each failure occurs with positive probability. Each failure increases d by 1. By independence, there is a positive probability of repeated failure without repairs, giving us the result.

Note: this result does not tell us where each of the states for the set equality will be (e.g. we may not have the state $(0, N) \in R$, but we will have a state with $d(s) = N$).
Note: this result means that there will be a state where the system is down, hence we will never have 100% uptime.

Discounted Linear Program

Maximise:

$$\sum_{s,a} r(s,a) x_{s,a}$$

Subject to:

$$\sum_{a \in A(j)} x_{j,a} - \gamma \sum_{s \in S} \sum_{a \in A(s)} P(j \mid s,a) x_{s,a} = \alpha_j \qquad \forall j \in S$$

$$x_{s,a} \geq 0 \qquad \forall a \in A(s)\ \forall s \in S$$

Optionally we can also restrict uptime $\geq U \in [0,1]$:

$$\sum_{s \in S} \sum_{a \in A(s)} x_{s,a} \left(\mathbb{1}_{\{s_1+s_2 \leq N-k\}} - U \right) \geq 0$$

α must be a stochastic vector that is strictly positive in each entry; we have picked $\alpha_j = \frac{1}{|S|}$

Undiscounted Linear Program

Maximise:

$$\sum_{s,a} r(s,a) x_{s,a}$$

Subject to:

$$\sum_{a \in A(j)} x_{j,a} - \sum_{s \in S} \sum_{a \in A(s)} P(j|s,a) x_{s,a} = 0 \qquad \forall j \in S$$

$$\sum_{a \in A(j)} x_{j,a} + \sum_{a \in A(j)} y_{j,a} - \sum_{s \in S} \sum_{a \in A(s)} P(j|s,a) y_{s,a} = \alpha_j \qquad \forall j \in S$$

$$x_{s,a} \geq 0, \; y_{s,a} \geq 0 \qquad \forall a \in A(s) \; \forall s \in S$$

Optionally we can also restrict uptime $\geq U \in [0,1]$:

$$\sum_{s \in S} \sum_{a \in A(s)} x_{s,a} \left(\mathbb{1}_{\{s_1 + s_2 \leq N-k\}} - U \right) \geq 0$$

Further extensions

- Consider the uptime cost of components i.e. $r(s, a)$ also incurs a cost $-s_0$
- Consider a finite time horizon model with system failure as an absorbing class
- Consider a model that chose k, N. Larger k means larger files sizes on each server $\implies$ larger repair cost and larger uptime cost.